

INTERNSHIP REPORT

ON MACHINE LEARNING APPLICATIONS IN INTELLIGENT TOLL MANAGEMENT SYSTEMS

Submitted To



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

IILM UNIVERSITY, GREATER NOIDA, U.P.

In partial fulfillment of the requirement for the degree of

B.TECH (CSE)

Submitted By

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Roll Number: 25SCS1003000457

Batch: 2026-27

Internship Organization

KENT Intelligent Transportation Systems (India) Pvt. Ltd.

Internship Period: 15 July 2026 – 15 August 2026

Certificate Reference: KENT/HR/290826

CANDIDATE'S DECLARATION

I, Dev Singh, hereby declare that the internship report titled “Machine Learning Applications in Intelligent Toll Management Systems” has been prepared for the academic requirements of the Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida.

The report documents the learning activities, technical concepts, practical exercises, and observations associated with my Machine Learning internship engagement at KENT Intelligent Transportation Systems (India) Pvt. Ltd. during the period from 15 July 2026 to 15 August 2026.

The report has been prepared using the internship certificate issued by the organization, the internship task framework used as a reference for the Machine Learning track, and technical documentation and background information relevant to the organization's toll-automation and intelligent transportation domain.

Where technical examples, datasets, model outputs, or diagrams are presented for explaining a learning task, they are intended as academic demonstrations of the workflow rather than representations of confidential KENT production data or proprietary source code.

I further declare that this report has been prepared for academic submission and that the sources used for technical explanations have been acknowledged in the references.

Signature: _____

Name: Dev Singh

Roll Number: 25SCS1003000457

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to KENT Intelligent Transportation Systems (India) Pvt. Ltd. for providing me with the opportunity to gain professional exposure to Machine Learning in an Intelligent Transportation Systems environment.

The internship period helped me understand how concepts studied in Computer Science and Engineering can be connected with real-world transportation technology. The exposure to toll automation, vehicle classification, traffic management, electronic toll collection, data analysis, and machine learning concepts helped me develop a broader view of intelligent systems.

I am thankful to IILM University, Greater Noida, and the School of Computer Science and Engineering for providing the academic environment and support required for completing the internship and preparing this report.

I am also grateful to the faculty members, mentors, colleagues, friends, and family members whose guidance and encouragement supported me throughout the learning process.

Finally, I appreciate the practical value of the internship because it encouraged me to approach machine learning as a complete workflow involving problem understanding, data preparation, analysis, model development, evaluation, documentation, and communication.

Dev Singh

B.Tech (CSE)

IILM University, Greater Noida

INTERNSHIP COMPLETION CERTIFICATE

The internship certificate issued by KENT Intelligent Transportation Systems (India) Pvt. Ltd. records that Dev Singh was associated with the organization from 15 July 2026 to 15 August 2026 and was involved in learning and contributing to activities related to Machine Learning.

The certificate further states that the conduct and performance during the period of association were satisfactory and wishes the student success in future endeavors.



Date: 29/08/2026
Ref: KENT/HR/290826

TO WHOMSOEVER IT MAY CONCERN

It is certified that **Mr. Dev Singh** was associated with us from **15th July 2026 to 15th August 2026**. During this period, he was involved in learning and contributing to activities related to **Machine Learning** as part of his professional engagement with the organization.

We found **Mr. Dev Singh** sincere, hardworking and committed towards his assigned responsibilities. His conduct and performance during the period of his association were found to be satisfactory.

We wish him all the best in his future endeavors.

Yours Truly,

For KENT Intelligent Transportation Systems (India) Pvt. Ltd.



AUTHORIZED SIGNITORY

Dated 29th August 2026

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1. PROJECT DESCRIPTION

Machine Learning is a branch of Artificial Intelligence in which algorithms learn useful patterns from data and use those patterns to make predictions, classifications, or other decisions. The importance of machine learning has increased because modern systems generate large amounts of structured and unstructured data that can be analyzed computationally.

The internship at KENT Intelligent Transportation Systems (India) Pvt. Ltd. was carried out in the Machine Learning domain from 15 July 2026 to 15 August 2026. KENT operates in the field of Intelligent Transportation Systems and toll automation. This makes the organization a useful context for studying how data-driven techniques can be connected with vehicle classification, traffic analysis, transaction monitoring, and intelligent transportation applications.

The purpose of this report is to present the internship learning as a systematic technical workflow. The report begins with the organization and its domain, then introduces Machine Learning concepts, describes the tools and basic tasks, and finally connects those tasks with intelligent toll and traffic-management scenarios.

KENT Intelligent Transportation Systems (India) Pvt. Ltd. is an Indian company established in 1998 and focused on toll automation and Intelligent Transportation Systems. The organization describes itself as a system integrator that designs solutions according to client requirements and local operational conditions.

KENT's publicly described offerings include automated toll collection, advanced traffic management, electronic toll collection, parking management, embedded traffic-management products, RFID products, toll lane controllers, automatic vehicle classification systems, and related equipment.

The organization also describes its Toll Xplore platform as a toll collection system designed to record transactions, support audit functions, and integrate with technologies such as automatic vehicle classification, electronic toll collection, and weigh-in-motion systems. These characteristics provide a practical context for understanding the value of data processing and machine learning.

The internship lasted one month. The learning approach followed a progression from basic Python and machine-learning concepts toward data preparation, exploratory analysis, model development, evaluation, and domain-oriented applications.

The task structure used in this report is intentionally based on beginner-to-intermediate Machine Learning activities. The tasks are suitable for a student internship and are connected to the transportation domain without claiming access to confidential production datasets.

Toll plazas and transportation systems produce continuous streams of information. Examples include vehicle categories, lane activity, transaction time, traffic volume, vehicle measurements, payment mode, and system events. Analyzing such information manually becomes difficult when the volume of data increases.

The problem addressed through the internship learning was to understand how a machine-learning workflow can process transportation-related data and support classification, prediction, and anomaly identification. The focus was on understanding the complete workflow rather than building a proprietary production system.

- Understand the fundamentals of Machine Learning.
- Develop practical Python skills for data analysis.
- Learn how transportation data can be represented as machine-learning features.
- Perform data cleaning and preprocessing.
- Carry out exploratory data analysis and visualization.
- Understand vehicle classification as a supervised-learning problem.
- Understand traffic-volume prediction as a regression problem.
- Study anomaly detection for unusual transportation observations.
- Train and evaluate basic machine-learning models.
- Compare models using appropriate performance metrics.
- Understand the connection between ML and intelligent toll systems.
- Improve documentation, debugging, and analytical problem-solving skills.

The report covers academic and practical machine-learning workflows relevant to toll automation and ITS. It does not disclose or reproduce proprietary KENT algorithms, confidential customer information, operational credentials, or private datasets. Demonstration examples use generic transportation concepts and illustrative data structures.

2. KENT AND INTELLIGENT TOLL AUTOMATION

Toll automation refers to the use of hardware and software to detect vehicles, classify them, process transactions, control lanes, collect payments, and maintain records with reduced dependence on manual intervention. An automated toll environment may combine sensors, lane controllers, software, RFID systems, cameras, barriers, displays, and transaction databases.

KENT's product information describes its Toll Xplore automated toll collection system as a scalable and configurable solution. The system records transactions at toll booths and automatically classifies vehicles through an Automatic Vehicle Classifier. It can also integrate with electronic toll collection and weigh-in-motion systems.

Toll Xplore is described by KENT as a toll collection software platform with strong audit capabilities. It records toll-booth transactions and maintains automatic vehicle-classification information. In cases where a toll collector's declaration differs from the AVC classification, captured images can be associated with the vehicle record to help identify the situation.

From a machine-learning perspective, such an environment illustrates the importance of structured data, classification outputs, image information, audit trails, and anomaly analysis. The internship therefore used toll automation as a domain context for learning rather than attempting to reproduce the proprietary Toll Xplore software.

Electronic Toll Collection allows vehicles to pass through a tolling point while payment is processed electronically. KENT describes its ETC system as using communication between ETC antennas and RFID tags to enable automatic payment.

For machine learning, ETC transaction records can potentially support analysis of traffic patterns, transaction behavior, lane utilization, and operational anomalies. Any such use requires appropriate data governance, privacy controls, and system-level validation.

Vehicle classification is particularly relevant to toll collection because toll charges may depend on vehicle category. KENT's AVC documentation describes classification based on vehicle profile information including vehicle length, maximum height, height variation, and number of axles, axle placement, and other discrimination criteria.

KENT also describes its AVC system as generating a vehicle profile using laser profiler sensors and using pattern-recognition methods to produce a vehicle class output. This provides a natural conceptual connection to machine learning classification, where measurable features are mapped to predefined classes.

KENT's Advanced Traffic Management System is described as a real-time system intended to provide information about road, traffic, and weather conditions. Its described subsystems include traffic control centers, emergency call boxes, variable message signs, automatic traffic counters and classifiers, weather monitoring, CCTV, incident detection, and management information systems.

These systems demonstrate why transportation organizations benefit from data analysis. Machine learning can complement conventional rule-based systems by identifying patterns in historical and real-time observations.

3. MACHINE LEARNING FUNDAMENTALS

Artificial Intelligence is the broader field concerned with creating systems capable of performing tasks associated with intelligent behavior. Machine Learning is a subset in which systems learn patterns from examples or data.

Supervised learning uses labeled data. Unsupervised learning works without predefined labels and attempts to identify structure. Reinforcement learning uses rewards and interaction to learn a policy. The internship focused primarily on supervised and unsupervised learning because they are appropriate for introductory data-driven classification, prediction, clustering, and anomaly-analysis tasks.

In supervised learning, each training example consists of input features and a target value. For classification, the target is categorical. For regression, the target is numerical.

Problem	Example Transportation Target	Typical Algorithms
Classification	Vehicle class	Decision Tree, Random Forest, KNN, Logistic Regression
Regression	Traffic volume	Linear Regression, Decision Tree Regressor, Random Forest Regressor

Unsupervised learning can be used when the objective is to discover groups or unusual patterns. K-Means clustering is a common introductory algorithm. Anomaly detection can also be approached using statistical methods or algorithms such as Isolation Forest.

A practical ML lifecycle can be represented as: Problem Definition → Data Collection → Data Cleaning → EDA → Feature Engineering → Train/Test Split → Model Training → Validation → Testing → Evaluation → Documentation.

Classification can be evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Regression can be evaluated using MAE, MSE, RMSE, and R-squared. The correct metric depends on the application and the cost of different types of errors.

4. TECHNOLOGIES AND TOOLS

Python was treated as the primary programming language because it provides a readable syntax and a mature ecosystem for data analysis and machine learning.

NumPy provides numerical arrays and mathematical operations used throughout scientific computing.

Pandas provides DataFrame-based tools for loading, inspecting, cleaning, transforming, grouping, and summarizing tabular datasets.

Matplotlib supports data visualization through line charts, bar charts, histograms, scatter plots, and other figures.

Scikit-learn provides classical machine-learning algorithms, preprocessing utilities, train-test splitting, cross-validation, and evaluation metrics.

Jupyter is useful for interactive experiments and visual analysis, while VS Code provides a structured development environment for Python projects.

Version control and documentation help preserve changes, organize experiments, and make technical work easier to reproduce.

```
python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

The above imports represent a typical beginner machine-learning environment. Actual libraries depend on the project requirements.

5. INTERNSHIP TASK 1 – MACHINE LEARNING ENVIRONMENT SETUP

The first task was to understand how to prepare a Python environment for data analysis and machine learning. A correctly configured environment is important because machine-learning projects often depend on several libraries that must work together.

1. Install or verify Python.
2. Create a project folder and organize files.
3. Install NumPy, Pandas, Matplotlib and Scikit-learn.
4. Open the project in Jupyter Notebook or VS Code.
5. Run basic import statements.
6. Verify that sample data can be loaded.
7. Document the packages and their purposes.

The task demonstrated that software setup is part of machine-learning development. Dependency problems can prevent otherwise correct code from running, so maintaining a consistent environment is important.

```
python
import sklearn
import pandas as pd
import numpy as np

print("Pandas:", pd.__version__)
print("NumPy:", np.__version__)
print("Scikit-learn:", sklearn.__version__)
```

The expected outcome of this task was a working environment capable of supporting the remaining internship exercises.

6. INTERNSHIP TASK 2 – DATA UNDERSTANDING AND COLLECTION

The second task focused on understanding how a transportation dataset can be structured for machine learning. A toll-related dataset can contain one record per vehicle transaction or one record per observation generated by a traffic sensor.

Feature	Type	Purpose
Vehicle_Length	Numeric	Represents measured vehicle length
Vehicle_Height	Numeric	Represents vehicle height
Axle_Count	Numeric	Number of detected axles
Lane_ID	Categorical	Identifies the lane
Hour	Numeric	Hour of observation
Vehicle_Class	Categorical	Target class for classification
Traffic_Volume	Numeric	Number of vehicles in a time interval
Payment_Mode	Categorical	Example transaction category

```
python
df = pd.read_csv("traffic_data.csv")
print(df.head())
print(df.info())
print(df.describe(include="all"))
```

The dataset should be inspected for missing values, duplicate records, invalid measurements, inconsistent category names, and incorrect data types. These checks should be performed before model training.

The exercise helped connect abstract ML features to toll-system concepts. Vehicle measurements can be used for classification, while time and lane variables can be useful for traffic analysis.

7. INTERNSHIP TASK 3 – DATA PREPROCESSING

Data preprocessing transforms raw data into a form suitable for analysis and machine learning. It is one of the most important steps because a model can only learn effectively from meaningful and consistent input.

Missing values can be handled by removing unsuitable records or imputing values using an appropriate strategy. The choice depends on why values are missing and how much information would be lost.

Duplicate records can distort analysis and may cause a model to see the same example repeatedly. Invalid measurements, such as impossible negative vehicle dimensions, should be investigated and corrected or removed when justified.

Machine-learning algorithms generally require numerical representations. Categorical variables such as lane ID or payment mode can be converted using techniques such as one-hot encoding or ordinal encoding when appropriate.

```
from sklearn.preprocessing import OneHotEncoder

encoder = OneHotEncoder(handle_unknown="ignore")
encoded = encoder.fit_transform(df[["Lane_ID", "Payment_Mode"]])
```

Some algorithms are sensitive to the scale of numerical features. Standardization can transform features to a common scale, although tree-based algorithms generally do not require feature scaling.

The task resulted in a cleaner dataset and a clearer understanding of how preprocessing choices affect the machine-learning pipeline.

8. INTERNSHIP TASK 4 – EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was used to understand patterns before selecting a model. EDA combines numerical summaries and visualizations.

A bar chart can be used to examine how frequently each vehicle category occurs. An imbalanced dataset may affect classification performance and should be considered when interpreting accuracy.

Grouping observations by hour can reveal peak and non-peak periods. Such analysis is useful for understanding traffic flow and can support later prediction tasks.

```
hourly = df.groupby("Hour")["Traffic_Volume"].sum()
hourly.plot(kind="line")
plt.xlabel("Hour")
plt.ylabel("Traffic Volume")
plt.title("Traffic Volume by Hour")
plt.show()
```

Correlation analysis can provide an initial view of relationships between numerical variables. Correlation does not by itself prove causation, and non-linear relationships may not be captured by a simple correlation coefficient.

EDA provides a foundation for selecting features, identifying anomalies, understanding distributions, and deciding which modelling approaches are appropriate.

9. INTERNSHIP TASK 5 – VEHICLE CLASSIFICATION

Vehicle classification was selected as a domain-relevant supervised-learning exercise. KENT's publicly described AVC technology classifies vehicles using profile characteristics such as length, height, height variation, number of axles, and axle placement. The internship exercise uses these concepts as a learning context.

The task can be expressed as learning a function that maps vehicle features to a predefined vehicle class. For example, the classes may include car, bus, light commercial vehicle, truck, or other predefined categories.

Feature	Example Role
Length	Separates compact and long vehicles
Height	Helps distinguish vehicle types
Axle Count	Useful for heavy-vehicle categories
Axle Placement	Adds structural information
Height Variation	Describes profile shape

A Decision Tree can split the data using feature thresholds. It is easy to visualize and explain, making it appropriate for an introductory classification task.

```
from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
pred = model.predict(X_test)
```

Random Forest combines multiple decision trees and averages or votes across them. It can provide stronger generalization than a single tree in many datasets.

The main learning outcome was understanding how measurable vehicle characteristics can be converted into model features and mapped to vehicle classes. The exercise also highlighted the importance of labelled training data.

10. INTERNSHIP TASK 6 – TRAFFIC VOLUME PREDICTION

Traffic-volume prediction was studied as a regression problem. The aim is to estimate the number of vehicles expected during a selected time interval using historical and contextual features.

- Hour of day
- Day of week
- Lane identifier
- Historical traffic volume
- Holiday indicator
- Weather-related variables, where available

Linear Regression models the relationship between input variables and a numerical target. It provides a simple baseline that is useful for understanding the regression workflow.

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
prediction = model.predict(X_test)
```

For regression, Mean Absolute Error represents the average magnitude of prediction errors, while Root Mean Squared Error gives greater weight to larger errors. R-squared indicates the proportion of variance explained by the model under the assumptions of the metric.

Traffic prediction can potentially support queue management, staffing decisions, lane planning, and traffic information. In a real deployment, the model would require continuous monitoring because traffic behaviour changes with events, road works, weather, and other conditions.

11. INTERNSHIP TASK 7 – ANOMALY DETECTION

—

Anomaly detection focuses on identifying observations that differ substantially from expected patterns. In toll and traffic systems, examples of unusual observations could include unexpected traffic volume, unusual transaction timing, or sensor measurements outside normal operating ranges.

—

A simple approach is to establish normal ranges using historical data. Observations far outside those ranges can be flagged for review. This method is easy to explain but may not handle complex relationships between variables.

—

Isolation Forest is an unsupervised anomaly-detection method that isolates observations through random partitioning. Observations that are easier to isolate tend to be considered more anomalous.

```
from sklearn.ensemble import IsolationForest

detector = IsolationForest(contamination=0.05, random_state=42)
labels = detector.fit_predict(X)
```

—

Anomaly detection can be considered for operational analytics, data-quality monitoring, or identifying records that require manual review. It should not automatically be treated as proof of fraud or wrongdoing; flagged observations require contextual verification.

—

The exercise demonstrated how unsupervised methods can complement supervised prediction and classification in a transportation analytics workflow.

12. INTERNSHIP TASK 8 – MODEL EVALUATION AND COMPARISON

The final core task focused on evaluating different approaches and understanding why a model should not be selected only because it produces a high score on one metric.

Metric	Meaning
Accuracy	Fraction of predictions that are correct
Precision	Fraction of predicted positives that are actually positive
Recall	Fraction of actual positives correctly detected
F1-score	Harmonic mean of precision and recall

A confusion matrix shows correct and incorrect predictions for each class. For vehicle classification, it can reveal whether one class is frequently confused with another.

Metric	Purpose
MAE	Average absolute prediction error
MSE	Average squared prediction error
RMSE	Square root of MSE; emphasizes larger errors
R^2	Measures explained variance relative to a baseline

Approach	Strength	Limitation
Logistic Regression	Simple and interpretable	Limited for complex boundaries
Decision Tree	Easy to explain	Can overfit
Random Forest	Robust ensemble method	Less interpretable than one tree
KNN	Simple concept	Sensitive to scaling and data size

The task strengthened the understanding that model selection should consider accuracy, computational requirements, interpretability, robustness, and the actual business or operational objective.

13. INTEGRATED ML WORKFLOW FOR TOLL SYSTEMS

A conceptual machine-learning-enabled toll analytics workflow can be represented as:

Vehicle / Sensor Data → Data Acquisition → Data Validation → Preprocessing → Feature Extraction → ML Model → Classification / Prediction / Anomaly Score → Analytics Dashboard / Review

Transportation systems may receive information from vehicle detectors, automatic vehicle classifiers, RFID/ETC systems, cameras, lane controllers, and other sensors. KENT's product information describes several of these components as part of its toll and traffic-management ecosystem.

The processing layer validates records, handles missing information, transforms features, and prepares the data for the selected model.

Different models can serve different tasks. Classification models can assign vehicle categories, regression models can estimate traffic volume, and anomaly-detection methods can flag unusual observations.

The final output may be a predicted value, a vehicle class, an anomaly score, or an analytical report. In real systems, the output should be monitored and verified according to operational requirements.

Machine learning should support operational decisions rather than being treated as infallible. Review mechanisms are especially important when an incorrect classification could affect toll charging or when an anomaly could trigger an operational response.

14. IMPLEMENTATION DETAILS

For academic exercises, a structured CSV dataset can be used. Each row represents an observation and each column represents a feature or target. The dataset should be separated into input features X and target y .

```
X = df[["Vehicle_Length", "Vehicle_Height", "Axle_Count"]]
y = df["Vehicle_Class"]
```

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
```

A fixed random state improves reproducibility of the split. Stratification can help preserve class proportions in classification problems when appropriate.

The selected model is fitted using training data. Preprocessing transformations should be learned from training data and then applied to test data to avoid leakage.

After training, the model generates predictions for unseen test observations. Predictions are compared with known target values to calculate evaluation metrics.

Each experiment should record the dataset version, features, preprocessing steps, model configuration, evaluation metrics, and observations. This makes it easier to reproduce results and compare experiments.

Reproducibility is particularly important in machine learning because small changes in preprocessing, random seeds, or feature selection can change results.

15. TESTING AND RESULTS

Testing was approached as a separate stage after model training. The objective is to evaluate how the learned model performs on data that was not used to fit its parameters.

For vehicle classification, the confusion matrix and class-wise precision, recall, and F1-score provide more information than accuracy alone. A balanced performance across classes is often more useful than a high overall score caused by a dominant class.

For traffic prediction, MAE and RMSE provide direct measures of prediction error. Results should be compared against a simple baseline to determine whether the model provides meaningful improvement.

Experiment	Metric	Result to Record	Interpretation
Vehicle Classifier	Accuracy	Insert actual value	Overall classification performance
Vehicle Classifier	F1-score	Insert actual value	Balance of precision and recall
Traffic Predictor	MAE	Insert actual value	Average absolute error
Traffic Predictor	RMSE	Insert actual value	Sensitivity to larger errors
Anomaly Detector	Flagged observations	Insert count	Requires contextual review

Because no proprietary KENT dataset or internship experiment log was provided for this report, numerical values have intentionally not been fabricated. The table is designed for insertion of actual results if they are available.

The most important result of the learning tasks is the understanding of the complete ML workflow and its relevance to transportation. The exercises demonstrate how raw observations can become structured features, how models can produce predictions, and how evaluation determines whether the approach is useful.

16. CHALLENGES FACED

Python projects may require several packages. Version incompatibilities or installation errors can prevent code from running. Keeping dependencies organized reduces these problems.

Transportation datasets can contain missing values, duplicates, sensor noise, inconsistent labels, and unexpected measurements. Data cleaning can require more effort than the model itself.

Including every available variable does not necessarily improve a model. Features should be selected based on relevance, availability, reliability, and the risk of data leakage.

A model may perform well on training data but poorly on unseen data. This can happen when a model learns noise or overly specific patterns.

Some vehicle classes may occur much more frequently than others. In such cases, accuracy may hide poor performance on minority classes.

A technically correct prediction may still require domain interpretation. For toll systems, a classification result may affect a transaction and therefore should be subject to system validation and operational rules.

Complex datasets and models may require additional memory and processing power. Efficient preprocessing and model selection help manage resource constraints.

17. LEARNING OUTCOMES

- Understanding of Machine Learning terminology and workflow.
 - Practical use of Python for data analysis.
 - Experience with tabular-data handling concepts.
 - Understanding of preprocessing and feature engineering.
 - Understanding of classification and regression.
 - Understanding of anomaly detection.
 - Experience with model evaluation concepts.
 - Understanding of data visualization.
 - Awareness of ML applications in transportation.
-

The internship improved the ability to look at a technical problem as a sequence of smaller stages. Instead of immediately selecting an algorithm, the workflow begins with understanding the problem and data, followed by preparation, modelling, evaluation, and interpretation.

- Improved technical documentation.
 - Improved debugging and problem-solving.
 - Better understanding of reproducible workflows.
 - Improved ability to explain technical concepts.
 - Greater awareness of data quality and validation.
 - Understanding of the importance of domain knowledge.
-

The most relevant domain outcome was understanding how machine learning can complement toll automation and intelligent transportation systems. Vehicle classification, traffic prediction, and anomaly detection provide clear examples of this connection.

18. APPLICATIONS AND FUTURE SCOPE

Machine learning can complement sensor-based vehicle profiling by learning relationships between measured vehicle characteristics and predefined classes. The approach can be evaluated using labelled examples and class-wise metrics.

Traffic prediction can help estimate future traffic conditions and may support operational planning. More advanced future work could include time-series models and deep learning.

Anomaly detection can be used to identify observations that differ from normal patterns. Such systems can help prioritize records for human review, although an anomaly should not automatically be interpreted as a confirmed fault or fraud.

Computer vision can support vehicle detection, counting, classification, licence-plate-related workflows, and incident monitoring. Future student projects could explore object-detection models using publicly available datasets.

Transportation systems increasingly require low-latency decisions. Running suitable models near sensors or lane equipment can reduce communication delays, although hardware limitations must be considered.

Explainable AI methods can help users understand why a model produced a particular result. This is useful when model outputs influence operational decisions.

Future work could combine real-time sensor streams, historical traffic data, model monitoring, dashboards, and automated alerts into a complete educational ITS analytics platform.

19. CONCLUSION

The Machine Learning internship at KENT Intelligent Transportation Systems (India) Pvt. Ltd. provided a useful context for studying the application of data-driven techniques to intelligent transportation and toll automation.

The report covered a sequence of introductory and practical tasks: environment setup, data understanding, preprocessing, exploratory analysis, vehicle classification, traffic-volume prediction, anomaly detection, and model evaluation. Together, these tasks represent the main stages of a basic machine-learning workflow.

The organization context made the learning more meaningful because toll automation involves vehicle detection, classification, transaction processing, electronic toll collection, traffic management, and data analysis. Machine learning can complement these systems by extracting patterns from transportation data and producing predictions or classifications.

A major learning outcome was that the success of a machine-learning solution depends not only on the algorithm but also on data quality, feature selection, evaluation methodology, system constraints, and domain understanding.

Overall, the internship strengthened the foundation required for further study in Machine Learning, Artificial Intelligence, Data Science, Computer Vision, and Intelligent Transportation Systems. It also improved practical understanding of how a technical problem can be converted into a structured and testable workflow.

20.1 DEMONSTRATION REFERENCE SCREENSHOTS

The following screenshots are academic reference outputs generated using a synthetic DEMO DATASET. They illustrate the machine-learning workflow described in this report and do not represent KENT proprietary systems, confidential data, or production results.

```
01 — Environment Setup

Python 3.x environment verified
> Import pandas as pd
> Import numpy as np
> import matplotlib.pyplot as plt
> from sklearn.model_selection import train_test_split
> from sklearn.metrics import accuracy_score
All required ML libraries imported successfully.

DEMO DATASET — illustrative academic reference only
```

Figure 20.1: Environment Setup — DEMO DATASET reference screenshot.

```
02 — Dataset Loading

> df = pd.read_csv('traffic_data.csv')
> df.head()

Vehicle_Length  Vehicle_Height  Axle_Count  Lane_ID  Hour  Vehicle_Class
          10.07           3.13         3    L3   23    Bus
           4.15           1.5         2    L3   14    Car
Rows: 180 | Columns: 8

DEMO DATASET — illustrative academic reference only
```

Figure 20.2: Dataset Loading — DEMO DATASET reference screenshot.

03 — Data Preprocessing

```
> df.isnull().sum()
Missing values: 0 across selected fields
> df = df.drop_duplicates()
> OneHotEncoder(handle_unknown='ignore')
> Standardization applied where algorithm requires scaling
Preprocessing complete: clean, encoded feature matrix prepared.
```

DEMO DATASET — illustrative academic reference only

Figure 20.3: Data Preprocessing — DEMO DATASET reference screenshot.

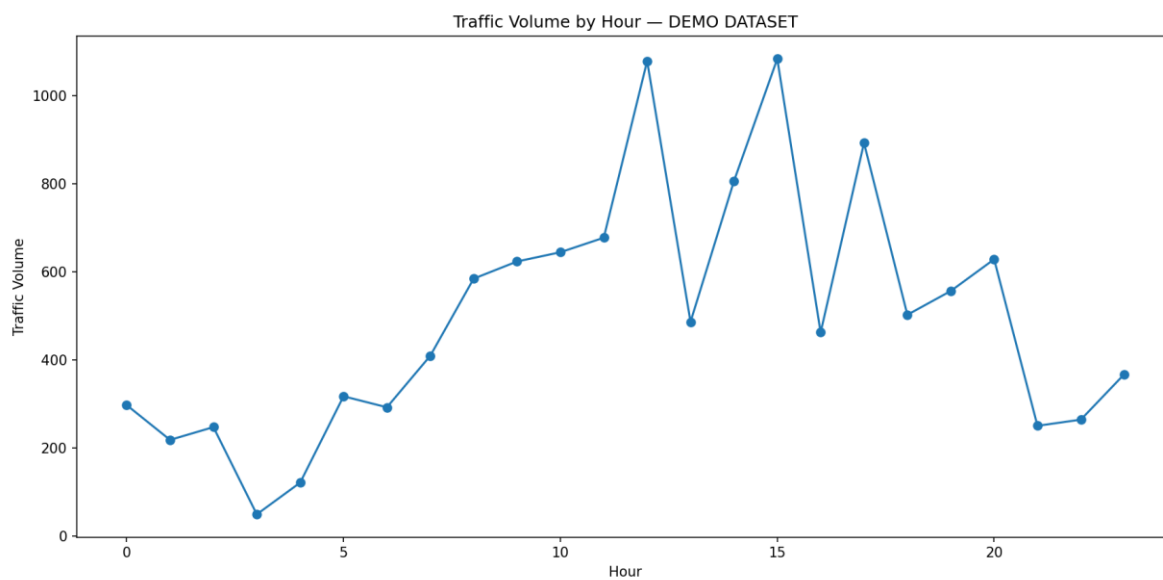


Figure 20.4: Exploratory Data Analysis — DEMO DATASET reference screenshot.

05 — Vehicle Classification

```
> X = df[['Vehicle_Length','Vehicle_Height','Axle_Count']]
> y = df['Vehicle_Class']
> model = DecisionTreeClassifier(random_state=42)
> model.fit(X_train, y_train)
> pred = model.predict(X_test)
Classification model trained successfully.
Classes: Car | LCV | Bus | Truck
```

DEMO DATASET — illustrative academic reference only

Figure 20.5: Vehicle Classification — DEMO DATASET reference screenshot.

06 — Traffic Volume Prediction

```
> model = LinearRegression()
> model.fit(X_train, y_train)
> prediction = model.predict(X_test)
Regression task completed on demonstration traffic-volume data.
Target: Traffic_Volume
Evaluation metrics: MAE, RMSE, R2
```

DEMO DATASET — illustrative academic reference only

Figure 20.6: Traffic Volume Prediction — DEMO DATASET reference screenshot.

07 — Anomaly Detection

```
> detector = IsolationForest(contamination=0.05, random_state=42)
> labels = detector.fit_predict(X)
```

Normal observations: 95% (approx.)
Flagged observations: ~5% (illustrative)
Flagged records require contextual review; they are not proof of fraud.

DEMO DATASET — illustrative academic reference only

Figure 20.7: Anomaly Detection — DEMO DATASET reference screenshot.

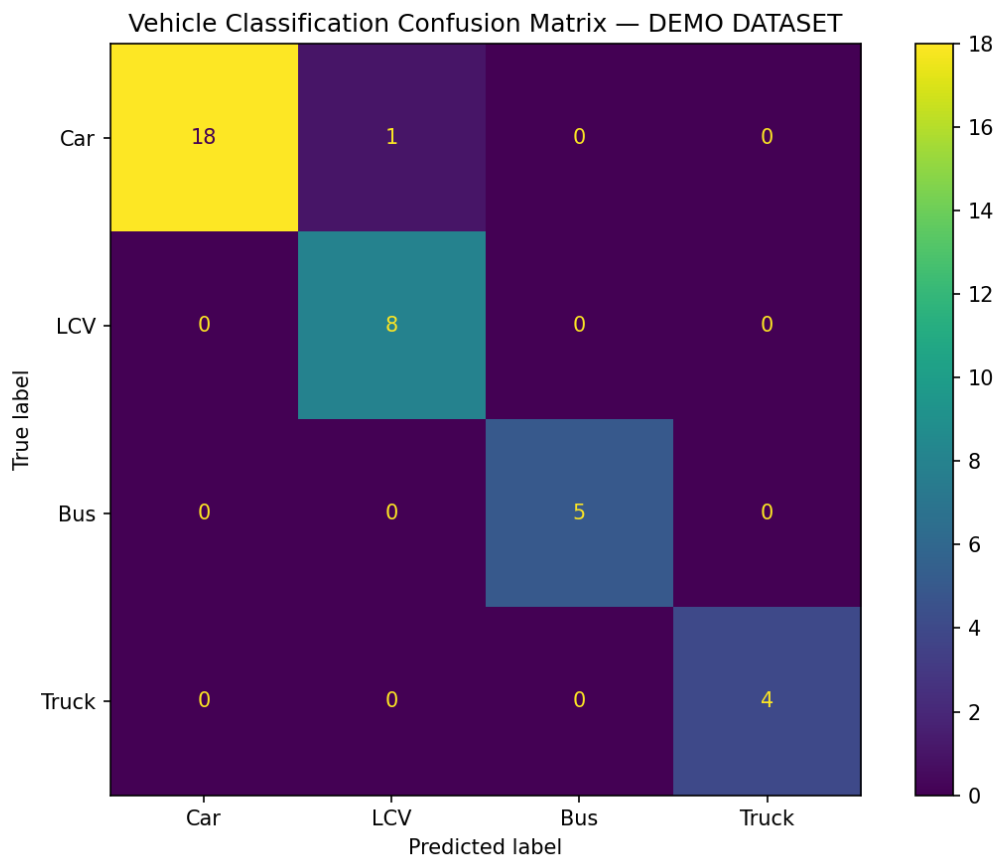


Figure 20.8: Confusion Matrix — DEMO DATASET reference screenshot.

09 — Integrated ML Pipeline

Vehicle / Sensor Data
↓
Data Acquisition → Validation → Preprocessing
↓
Feature Extraction → ML Model
↓
Classification / Prediction / Anomaly Score
↓
Analytics Dashboard / Human Review

DEMO DATASET — illustrative academic reference only

Figure 20.9: Integrated ML Pipeline — DEMO DATASET reference screenshot.

10 — Model Comparison

MODEL COMPARISON — DEMO WORKFLOW

Logistic Regression	Simple, interpretable	Complex boundaries may limit
Decision Tree	Easy to explain	Can overfit
Random Forest	Robust ensemble	Less interpretable
KNN	Simple concept	Sensitive to scaling/data size

Selection should consider metrics + operational requirements.

DEMO DATASET — illustrative academic reference only

Figure 20.10: Model Comparison — DEMO DATASET reference screenshot.